Boron and Mg/Ca Calibration details for Brabson et al., 2016 in prep

Species (300-425 µm)	**Label for Boron**	**Sample ID**	**Site**
Trilobatus trilobus	B1	12LB	292
Globorotalia tumida	B2	12LB	292
Sphaeroidinella dehiscens	B3	12LB	292
Globorotalia menardii	B4	12LB	292
Globorotalia crassaformis	B5	12LB	292
Neogloboquadrina dutertrei	B6	14LB	1210A
Globorotalia crassaformis	B7	14LB	1210A
Globorotalia truncatulinioides	B8	14LB	1210A
Uvigerina peregrina	B9	14LB	1210A
Globorotalia crassaformis	B10	16LB	1210A
Planulina wuellerstorfi	B11	16LB	1210A
Trilobatus sacculifer	B12	6LB	U1482A
Globigerinoides ruber	B13	6LB	U1482A
Trilobatus trilobus	B14	6LB	U1482A

Globorotalia tumida	B15	6LB	U1482A
Globorotalia ungulata	B16	6LB	U1482A
Globigerinella siphonifera	B17	6LB	U1482A
Pulleniatina obliquiloculata	B18	6LB	U1482A
Orbulina universa	B19	6LB	U1482A
Trilobatus sacculifer	B20	8LB	U1482A
Globigerinoides ruber	B21	8LB	U1482A
Trilobatus trilobus	B22	8LB	U1482A
Globorotalia menardii	B23	8LB	U1482A
Globorotalia crassaformis	B25	8LB	U1482A
Orbulina universa	B26	8LB	U1482A
Trilobatus sacculifer	B27	1LB	U1490C
Orbulina universa	B28	1LB	U1490C
Trilobatus trilobus	B29	1LB	U1490C
Globigerinella siphonifera	B30	1LB	U1490C
Globorotalia tumida	B31	1LB	U1490C
Globorotalia ungulata	B32	1LB	U1490C
Pulleniatina obliquiloculata	B33	1LB	U1490C

Trilobatus sacculifer	B34	4LB	U1490C
Globigerinoides ruber	B35	4LB	U1490C
Trilobatus trilobus	B36	4LB	U1490C
Globorotalia tumida	B37	4LB	U1490C
Globorotalia ungulata	B38	4LB	U1490C
Pulleniatina obliquiloculata	B39	4LB	U1490C
Orbulina universa	B40	4LB	U1490C

Core Details	Age (Mya)	Latitude
31-292-4R-2W, 20-21cm	3.2	15.86
31-292-4R-2W, 20-21cm	3.2	15.86
31-292-4R-2W, 20-21cm	3.2	15.86
31-292-4R-2W, 20-21cm	3.2	15.86
31-292-4R-2W, 20-21cm	3.2	15.86
198-1210A-1H-1W, 0-4cm	top	32.22
198-1210A-1H-1W, 0-4cm	top	32.22
198-1210A-1H-1W, 0-4cm	top	32.22
198-1210A-1H-1W, 0-4cm	top	32.22
198-1210A-6H-6W, 120-121cm	3.2	32.22
198-1210A-6H-6W, 120-121cm	3.2	32.22
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12

363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-1H-1W, 2-3cm	top	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1482A-16H-2W, 110-111cm	3.2	-15.12
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82
363-U1490C-1H-1W, 0-1cm	top	5.82

363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82
363-U1490C-6H-3W, 23-24 cm	3.2	5.82

Longitude	water depth (m)		d11B (TPB corrected)	2SD on d11B
124.65	2940		19.07	0.20
124.65	2940		16.47	0.08
124.65	2940		18.49	0.24
124.65	2940		16.69	0.19
124.65	2940		14.53	0.16
158.25	2573		16.01	0.23
158.25	2573		16.29	0.28
158.25	2573		15.91	0.05
158.25	2573		13.06	0.25
158.25	2573		14.76	0.19
158.25	2573		13.63	0.21
120.35	1468		19.20	0.23
120.35	1468		20.12	0.16
120.35	1468		19.78	0.11

120.35	1468		16.37	0.18
120.35	1468		16.05	0.14
120.35	1468		17.09	0.20
120.35	1468		15.80	0.14
120.35	1468		17.38	0.24
120.35	1468		19.18	0.15
120.35	1468		20.12	0.28
120.35	1468		19.42	0.27
120.35	1468		17.52	0.30
120.35	1468		15.43	0.15
120.35	1468		18.00	0.11
142.65	2341		19.40	0.15
142.65	2341		13.43	0.11
142.65	2341		19.76	0.08
142.65	2341		18.23	0.09
142.65	2341		17.12	0.13
142.65	2341		17.16	0.10
142.65	2341		17.05	0.04

142.65	2341		18.76	0.17
142.65	2341		19.24	0.10
142.65	2341		18.63	0.23
142.65	2341		16.03	0.12
142.65	2341		16.10	0.24
142.65	2341		16.04	0.26
142.65	2341		17.06	0.17

A	1SD on A	B	1SD on B	Reference
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.57	1.2	6.33	2.52	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.65	0.76	5.36	1.36	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.72	0.74	3.88	0.65	Guillermic et al., 2020; Foster et al., 2008
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
1	0	0	0	Rae et al., 2011 (no calibration necessary)
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
1	0	0	0	Rae et al., 2011 (no calibration necessary)
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.60	0.11	9.52	2.02	Henehan et al., 2013
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015

0.57	1.2	6.33	2.52	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.59	0.65	5.59	4.16	Guillermic et al., 2020; Henehan et al., 2016
0.95	0.16	-0.42	2.85	Henehan et al., 2016
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.60	0.11	9.52	2.02	Henehan et al., 2013
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.65	0.76	5.36	1.36	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.95	0.16	-0.42	2.85	Henehan et al., 2016
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.95	0.16	-0.42	2.85	Henehan et al., 2016
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.57	1.2	6.33	2.52	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.59	0.65	5.59	4.16	Guillermic et al., 2020; Henehan et al., 2016

0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.60	0.11	9.52	2.02	Henehan et al., 2013
0.82	0.22	3.94	4.02	Sanyal et al., 2000 refit by Martinez-Boti et al., 2015
0.57	1.2	6.33	2.52	Guillermic et al., 2020
0.95	0.13	0.18	0.6	"Deep Dwellers" Guillermic et al., 2020; Foster et al., 2008; Henehan et al., 2016; Raitzsch et al., 2018
0.59	0.65	5.59	4.16	Guillermic et al., 2020; Henehan et al., 2016
0.95	0.16	-0.42	2.85	Henehan et al., 2016

d11B_borate (Species Corrected)	2SD on d11B	pH	2SD error on pH
18.45	0.20	8.07	0.04
17.79	0.08	8.07	0.02
19.27	0.24	8.23	0.04
17.43	0.19	8.01	0.04
15.11	0.16	7.76	0.05
16.85	0.23	8.05	0.05
16.96	0.28	8.07	0.06
16.56	0.05	8.03	0.01
13.06	0.25	7.43	0.19
15.35	0.19	7.89	0.06
13.63	0.21	7.65	0.12
18.61	0.23	8.07	0.04
17.67	0.16	7.98	0.03
19.32	0.11	8.13	0.02

17.61	0.18	8.05	0.04	
16.71	0.14	8.02	0.03	
17.80	0.20	8.02	0.04	
17.31	0.14	8.00	0.03	
18.74	0.24	8.06	0.04	
18.59	0.15	8.06	0.03	
17.67	0.28	7.97	0.05	
18.88	0.27	8.08	0.05	
18.71	0.30	8.10	0.05	
16.05	0.15	7.86	0.04	
19.39	0.11	8.15	0.02	
18.85	0.15	8.09	0.03	
14.58	0.11	7.57	0.04	
19.29	0.08	8.12	0.01	
19.00	0.09	8.13	0.02	
18.93	0.13	8.17	0.02	
17.87	0.10	8.15	0.02	
19.42	0.04	8.18	0.01	

18.07	0.17	8.02	0.03	
16.20	0.10	7.84	0.02	
17.91	0.23	8.00	0.04	
17.02	0.12	7.97	0.03	
16.76	0.24	8.01	0.05	
17.71	0.26	8.02	0.05	
18.40	0.17	8.10	0.03	

Mg24/Ca	A	1SD on A	B	1SD on B	Reference
3.43	0.090	0.000	0.347	0.011	Anand et al., 2003
2.34	0.090	0.000	0.380	0.000	Anand et al., 2003
1.90	0.090	0.000	0.380	0.000	Anand et al., 2003 (multi-species)
2.74	0.091	0.000	0.360	0.000	Regenberg et al., 2009
1.90	0.090	0.000	0.339	0.029	Anand et al., 2003
1.24	0.090	0.000	0.342	0.012	Anand et al., 2003
1.13	0.090	0.000	0.339	0.029	Anand et al., 2003
1.16	0.090	0.000	0.359	0.008	Anand et al., 2003
1.09	0.061	0.000	0.924	0.000	Lear et al., 2002
0.99	0.090	0.000	0.339	0.029	Anand et al., 2003
1.17	0.109	0.007	0.867	0.049	Lear et al., 2002
3.84	0.090	0.000	0.347	0.011	Anand et al., 2003
4.96	0.085	0.006	0.480	0.070	Anand et al., 2003
3.91	0.090	0.000	0.347	0.011	Anand et al., 2003

2.37	0.090	0.000	0.380	0.000	Anand et al., 2003
2.97	0.083	0.000	0.840	0.000	Anand et al., 2003 (deep dweller)
3.52	0.090	0.000	0.380	0.000	Anand et al., 2003 (multi-species)
2.48	0.090	0.000	0.328	0.007	Anand et al., 2003
8.23	0.090	0.000	0.595	0.042	Anand et al., 2003
4.14	0.090	0.000	0.347	0.011	Anand et al., 2003
5.25	0.085	0.006	0.480	0.070	Anand et al., 2003
4.44	0.090	0.000	0.347	0.011	Anand et al., 2003
3.56	0.091	0.000	0.360	0.000	Regenberg et al., 2009
2.57	0.090	0.000	0.339	0.029	Anand et al., 2003
6.03	0.090	0.000	0.595	0.042	Anand et al., 2003
4.06	0.090	0.000	0.347	0.011	Anand et al., 2003
6.88	0.090	0.000	0.595	0.042	Anand et al., 2003
4.12	0.090	0.000	0.347	0.011	Anand et al., 2003
3.42	0.090	0.000	0.380	0.000	Anand et al., 2003 (multi-species)
2.36	0.090	0.000	0.380	0.000	Anand et al., 2003
2.84	0.083	0.000	0.840	0.000	Anand et al., 2003 (deep dweller)
2.66	0.090	0.000	0.328	0.007	Anand et al., 2003

3.95	0.090	0.000	0.347	0.011	Anand et al., 2003
4.23	0.085	0.006	0.480	0.070	Anand et al., 2003
3.98	0.090	0.000	0.347	0.011	Anand et al., 2003
2.83	0.090	0.000	0.380	0.000	Anand et al., 2003
3.42	0.083	0.000	0.840	0.000	Anand et al., 2003 (deep dweller)
2.96	0.090	0.000	0.328	0.007	Anand et al., 2003
4.51	0.090	0.000	0.595	0.042	Anand et al., 2003

TEMP_C (Mg/Ca)	1SD on Temp
25.46	0.04
20.20	0.00
17.88	0.00
22.30	0.00
19.15	0.11
14.31	0.05
13.38	0.11
13.03	0.03
2.71	0.00
11.91	0.11
2.75	0.43
26.71	0.04
27.47	1.98
26.91	0.04

20.34	0.00
15.22	0.00
24.73	0.00
22.48	0.03
29.19	0.28
27.55	0.04
28.14	2.03
28.32	0.04
25.18	0.00
22.51	0.11
25.73	0.28
27.33	0.04
27.20	0.28
27.49	0.04
24.41	0.00
20.29	0.00
14.68	0.00
23.26	0.03

27.02	0.04
25.60	1.85
27.11	0.04
22.31	0.00
16.92	0.00
24.44	0.03
22.51	0.28

Region	Species	Leg	Site
NW PAC	C. wuellerstorfi	31	296
NW PAC	N. dutertrei	350	1436
NW PAC	C. wuellerstorfi	X	X
SHATSKY	C. wuellerstorfi	198	1209A
SHATSKY	C. wuellerstorfi	198	1210B
WWP	C. wuellerstorfi	89	586
WWP	C. wuellerstorfi	363	U1490
WWP	C. wuellerstorfi	130	806
WWP	C. wuellerstorfi	130	806
EEP	C. wuellerstorfi		849
EEP	C. wuellerstorfi	138	849
EEP	G. sacculifer	138	851E
EEP	G. sacculifer	136	851B
EEP	N. dutertrei	201	1241C
EEP	N. dutertrei	136	851B
EEP	G. tumida	138	851E
EEP	G. tumida	136	851B
EEP	C. wuellerstorfi	138	846
EEP	C. wuellerstorfi	138	846

Core Details	LAT	LON	Water Depth (mbsf)	δ13C
31-296-11R-3,113-115	29.34	133.53	2920	-0.07
350-U1436C-1H-1, 0-2	32.39	140.37	1771	0.51
KT89-18 P4	32.09	133.54	2700	-0.12
1209B-6H-1, 95-97	32.65	158.51	2387	0.03
1210B-6H-5, 145-147	32.22	158.26	2574	0.03
89-586A-4-5,50-52	-0.50	158.50	2207	-0.03
top	5.49	142.39	2341	0.07
top	0.32	159.36	2520	0.27
synthesis	0.32	159.36	2520	0.16
core depth 996'	-0.18	249.48	3839	0.27
top	0.18	249.47	3839	0.16
138-851E-6H-4,85-87	2.77	249.43	3760	2.83
851B,1H-1,10-15	2.77	249.43	3761	2.20
5H-4, 60-61	5.84	273.56	2027	0.89
851B,1H-1,10-15	2.77	249.43	3761	1.31
138-851E-6H-4,85-87	2.77	249.43	3760	1.67
851B,1H-1,10-15	2.77	249.43	3761	1.77
	-3.1	269.18	3296	-0.16
top	-3.1	269.18	3296	0.08

δ18O	Age (ka)
2.33	3180
-0.32	0
X	0
3.40	3197.5
3.41	3198.6
2.23	3180
3.19	0
X	0
X	mPWP
3.15	average of 319909 and 3203445
X	0
-1.49	3202
-0.91	0
-0.80	3200
0.72	0
0.21	3202
0.65	0
X	mPWP
X	0

Reference
Matsui et al., 2020
Vautravers et al., 2016
Oba and Yasuda 1992 - original - used in Matsumoto et al., 2002 ht
Ford et al., (2022)
Ford et al., (2022)
Whitman and Berger, 1993
BoscoloGalazzo_et_al., 2021
Bickert et al., 1993 https://doi.org/10.2973/odp.proc.sr.130.024.19
Ford et al., (2022)
Mix et al., 2005
Mix et al., 1995 https://doi.org/10.2973/odp.proc.sr.138.120.1995
Cannariato and Ravelo, 1997
Cannariato and Ravelo, 1997
Steph et al., 2005
Cannariato and Ravelo, 1997
Cannariato and Ravelo, 1997
Cannariato and Ravelo, 1997
Shackleton, 1995 from Ford et al., 2022
Mix et al., 1995 https://doi.org/10.2973/odp.proc.sr.138.120.1995

Notes

https://publications.iodp.org/proceedings/350/201/350_201.html

From compilation at Nederbragt, 2025

C. wuellerstorfi (>250 mm)

From compilation at Nederbragt, 2024

C. wuellerstorfi and peregrina

G. sacculifer (355-425 mm)

G. tumida (355-425 mm)

From compilation at Nederbragt, 2023 (noted use of PDB not VPDB, notes no offset for carbon)

From compilation at Nederbragt, 2023 (noted use of PDB not VPDB, notes no offset for carbon)

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From Shankle et al., 2021

Age	Depth	pH West (806)	ph East (846)
Modern	25-50m	8.07 ± 0.01	8.01 ± 0.06
3.000 Ma	50-100m (O. universa)	8.03 ± 0.05	7.83 ± 0.06